Unit: Unit 7: Factoring
Topic: Factoring by Grouping

Name: ____________________
Date: _____________

Foundation (Jalapeno)

Q1. Factor $x^2 + 3x + 2x + 6$ by grouping.
(A) $x(x+3) + 2(x+3)$
(B) $x^2 + 5x + 6$
(C) $x(x+2) + 3(x+2)$
(D) $x^2 + 2x + 3x + 6$
(E) $x(x+5) + 6$

Q2. A basketball player scored $3x + 2$ points in the first quarter and $2x - 5$ points in the second quarter. Factor the expression $3x + 2 + 2x - 5$ by grouping.
(A) $5x - 3$
(B) $x(5x - 3)$
(C) $x(3+2) - 5 + 2$
(D) $5x - 3$
(E) $x(2+3) - 5 + 2$

Q3. A cyclist traveled $x^2 + 4x$ miles in the first hour and $3x^2 - x$ miles in the second hour. Factor the expression $x^2 + 4x + 3x^2 - x$ by grouping.
(A) $x^2(1+3) + x(4-1)$
(B) $4x^2 + 3x$
(C) $x^2(1+3) + x(1-4)$
(D) $x(x+4) + 3x(x-1)$
(E) $x(x+1) + 3x(x+4)$

Q4. Factor $x^2 - 2x + 5x - 10$ by grouping.
(A) $x(x-2) + 5(x-2)$
(B) $x(x+5) - 2(x+5)$
(C) $x^2 + 3x - 10$
(D) $x(x+3) - 2(x+5)$
(E) $x(x-2) + 5(x+2)$

Q5. A football team scored $2x+5$ points in the first half and $x-3$ points in the second half. Factor the expression $2x+5+x-3$ by grouping.
(A) $3x + 2$
(B) $x(2+1) + 5 - 3$
(C) $2x + x + 5 - 3$
(D) $x(1+2) - 3 + 5$
(E) $x(2+1) + 2$

Q6. Factor $x^2 + 2x - 5x - 10$ by grouping.
(A) $x(x-3) - 2(x+5)$
(B) $x^2 - 3x - 10$
(C) $x(x+2) - 5(x+2)$
(D) $x(x-5) - 2(x+5)$
(E) $x(x+3) - 2(x+5)$

Q7. A tennis player served x^2+x aces in the first set and $2x-3$ aces in the second set. Factor the expression $x^2+x+2x-3$ by grouping.
(A) $x^2 + 3x - 3$
(B) $x(x+1) + 2(x-3)$
(C) $x(x+1) + 2(x+3)$
(D) $x(x+3) - 3$
(E) $x(x+2) + 3$

Q8. Factor $3x^2 + 2x - 6x - 4$ by grouping.
(A) $3x(x-2) - 2(x+2)$
(B) $x(3x+2) - 2(3x+2)$
(C) $3x(x+2) - 2(x+2)$
(D) $x(2+3) - 2(x+2)$
(E) $3x(x+2) + 2(x-2)$

Open Ended Questions (FRQ)

Q9. A team of athletes ran $x^2 + 4x$ miles on Monday and $2x^2 - 3x$ miles on Tuesday. Factor the expression $x^2 + 4x + 2x^2 - 3x$ by grouping and verify the result by multiplication.

Q10. Factor the expression $2x^2 + 5x - 3x - 15$ by grouping and then multiply the factors to verify the result.

Chef's Tips

- Provide clear instructions and examples before the test.
- Encourage students to use diagrams and algebraic manipulations to verify their answers.
- Remind students to check their work carefully before submitting their tests.